column-hse-z: hydrostatic columns on a geometric-height grid

Physics and method of `run.mode = "column-hse-z"` in HSEMHS. A self-contained description.

Abstract

Given the temperature of a three-dimensional piece of the solar atmosphere on a grid of geometric heights, this mode computes the gas pressure, the density, the electron density, the 500 nm extinction and the optical depth at every grid point. It does so by treating every vertical column of the grid as an isolated, static column of gas that carries its own weight (hydrostatic equilibrium), with the same prescribed pressure at one level of every column. The equation of state includes ionisation and molecules. The equation solved, its boundary condition, the numerical integration, the accuracy control, the computation of the optical depth and the assumptions are derived and explained step by step, as well as what the result does not satisfy: horizontal force balance.

1. Introduction

A radiative magnetohydrodynamic simulation or a model atmosphere describes the gas by its temperature T, its density ρ, its pressure P and its velocity. If only the temperature is kept, the simplest physically consistent way to recover the pressure and the density is to assume that the gas is *at rest* and that each vertical column supports its own weight. This mode implements exactly that. Its output serves two purposes: it is a stand-alone hydrostatic atmosphere with an optical-depth scale, and it is the starting point of the magnetohydrostatic modes, which add the horizontal force balance and the magnetic field.

Input: the temperature $T(x, y, z)$ on a rectangular grid, the coordinates x, y, z, the gravity g and one reference pressure P_{ref}. **Output:** P, ρ, n_e, P_e, $\bar{\mu}$, n_{H}, the column mass m, χ_{500} and τ_{500} on the same grid, the height of $\tau_{500} = 1$ in every column, and a diagnostic of the horizontal and vertical force imbalance.

2. Notation

Symbol	Meaning	Unit	Role
$x,\ y$	horizontal coordinates; grid x_i, $i = 1 \dots N_x$, y_j, $j = 1 \dots N_y$, uniform spacing Δx, Δy	m	given
z	height, growing upward; grid z_k, $k = 1 \dots N_z$, stored top first ($z_1 > z_2 > \cdots > z_{N_z}$)	m	given
T	temperature	K	given, fixed
X	$\ln(T/1\,\mathrm{K})$	–	given
P	gas pressure	Pa	unknown
u	$\ln(P/1\,\mathrm{Pa})$, the variable that is integrated	–	unknown
ρ	mass density, $\rho(T, P)$ from the equation of state	$\mathrm{kg\,m^{-3}}$	derived
q	ρ/P	$\mathrm{s^2\,m^{-2}}$	derived
$n_e,\ P_e$	electron number density; electron pressure $n_e k_B T$	$\mathrm{m^{-3}}$, Pa	derived
$\bar{\mu}$	mean molecular weight, $\rho k_B T/(P m_{\mathrm{u}})$	–	derived
n_{H}	number density of hydrogen nuclei, ρ divided by the mass per H nucleus	$\mathrm{m^{-3}}$	derived
H	pressure scale height $P/(\rho g)$	m	derived
m	column mass: mass above a point per unit horizontal area	$\mathrm{kg\,m^{-2}}$	derived
g	gravitational acceleration ($274\,\mathrm{m\,s^{-2}}$ at the solar surface)	$\mathrm{m\,s^{-2}}$	parameter
P_{ref}	pressure prescribed at the anchor level of every column	Pa	parameter
z_{ref}	anchor height (only for the anchor option `height`)	m	parameter
χ_{500}	extinction coefficient at 500 nm (absorption plus scattering)	$\mathrm{m^{-1}}$	derived
τ_{500}	optical depth at 500 nm along the vertical, counted downward from the top	–	derived
τ_{top}	optical depth of the gas above the top of the grid	–	derived
H_χ	scale height of χ_{500} at the top of the grid	m	derived
μ	cosine of the viewing angle (default 1, vertical)	–	parameter
$k_B,\ m_{\mathrm{u}}$	Boltzmann constant $1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}}$; atomic mass unit $1.661 \times 10^{-27}\,\mathrm{kg}$		constants
$\mathbf{F}$	net force per unit volume	$\mathrm{N\,m^{-3}}$	diagnostic

3. The force balance of a static gas

3.1 All three components

A small volume of gas is at rest when the forces on it cancel. Without a magnetic field, two forces act per unit volume: the pressure force $-\nabla P$ (gas is pushed from high to low pressure) and gravity $-\rho g\,\hat{\mathbf{z}}$. The static momentum equation is therefore

$$\mathbf{F} \equiv -\nabla P - \rho\, g\,\hat{\mathbf{z}} = 0, \qquad \text{i.e.} \qquad -\frac{\partial P}{\partial x} = 0, \quad -\frac{\partial P}{\partial y} = 0, \quad -\frac{\partial P}{\partial z} - \rho g = 0. \tag{1}$$

This mode imposes *only the vertical component*, separately in every column. The two horizontal components are not imposed; Sect. 10 shows why they cannot be satisfied in general, and how large they remain is reported as a diagnostic.

3.2 Hydrostatic equilibrium of a column

Take a thin horizontal slab of area A between z and $z + dz$. The gas below pushes it up with the force $P(z)A$, the gas above pushes it down with $P(z + dz)A$, and gravity pulls it down with its weight $\rho\, A\, dz\, g$. Equilibrium means $P(z)A - P(z + dz)A - \rho g A\, dz = 0$. Dividing by $A\, dz$ and letting $dz \to 0$ gives the **hydrostatic equation**

$$\boxed{\frac{dP}{dz} = -\rho\, g} \tag{2}$$

which is the z component of Eq. (1): the pressure falls upward, fastest where the gas is densest.

4. The equation of state and the opacity

Every force balance in this document involves the density ρ, and every optical depth involves the extinction coefficient χ_{500}. Neither is an unknown of its own: at given temperature and pressure both are fixed by the microphysics of the gas. This section explains how.

4.1 The ideal gas

In the solar photosphere and chromosphere the particles interact only weakly, so the gas obeys the ideal-gas law. If n is the number of free particles (atoms, ions, molecules and electrons) per cubic metre,

$$P = n\, k_B T. \tag{3}$$

If the average mass of a free particle is $\bar{\mu}\, m_{\mathrm{u}}$ (with m_{u} the atomic mass unit and $\bar{\mu}$ the **mean molecular weight**), the mass density is $\rho = n\, \bar{\mu}\, m_{\mathrm{u}}$, and therefore

$$\rho = \frac{\bar{\mu}\, m_{\mathrm{u}}}{k_B}\, \frac{P}{T}. \tag{4}$$

At fixed $\bar{\mu}$ and T the density is proportional to the pressure.

4.2 Why $\bar{\mu}$ depends on T and P: ionisation and molecules

For neutral solar gas (about 91 % hydrogen and 9 % helium atoms by number, plus a small fraction of heavier elements) $\bar{\mu} \approx 1.3$. When a hydrogen atom is ionised it becomes two free particles, a proton and an electron, of almost the same total mass; in fully ionised gas $\bar{\mu} \approx 0.6$. In the coolest layers hydrogen partly forms $\mathrm{H_2}$ molecules, which raises $\bar{\mu}$. The degree of ionisation of an element in stage i is given by the **Saha equation**

$$\frac{n_{i+1}\, n_e}{n_i} = \frac{2U_{i+1}(T)}{U_i(T)} \left(\frac{2\pi m_e k_B T}{h^2}\right)^{3/2} e^{-E_i/k_B T}, \tag{5}$$

with E_i the ionisation energy (13.6 eV for hydrogen), U_i the partition functions, m_e the electron mass and h Planck's constant. Two properties matter for us. (i) Because of the exponential, ionisation switches on over a narrow range of temperature. (ii) The electron density n_e appears on the left: at higher density there are more electrons to recombine with, and a higher temperature is needed for the same ionisation. Hence $\bar{\mu} = \bar{\mu}(T, P)$, and Eq. (4) is no longer a simple proportionality. The Saha equations of all elements are coupled through the common n_e and through charge conservation (n_e equals the total charge of the ions), and the molecular equilibria ($\mathrm{H_2}$, $\mathrm{H_2^+}$, $\mathrm{H^-}$) are solved together with them. This assumes **local thermodynamic equilibrium** (LTE): the level populations and ionisation stages are those of a gas in equilibrium at the local T.

4.3 The 500 nm extinction

Light of wavelength 500 nm travelling a distance ds loses the fraction $\chi_{500}\, ds$ of its intensity, by absorption and by scattering. In the solar photosphere χ_{500} is dominated by the negative hydrogen ion $\mathrm{H^-}$ (a hydrogen atom

with a second, weakly bound electron, binding energy 0.75 eV), whose abundance is proportional to the electron density. The electrons come mostly from easily ionised metals (Na, Mg, Si, Ca, Fe, ...), so χ_{500} grows steeply with temperature and with density. The code adds the continuum contributions of H (bound–free and free–free), H^-, H_2^+, He, He^-, the metals C, N, Mg, Si, Ca and Fe, Rayleigh scattering on H, He and H_2, and Thomson scattering on free electrons, each counted once. Spectral lines are not included: χ_{500} is a continuum extinction.

4.4 The table and its interpolation

Solving the coupled Saha, molecular and charge equations at every grid point at every step would be far too slow. They are therefore solved once, with Wittmann's equation-of-state code for solar composition (with a dedicated hot branch above a switch temperature $T_{\rm sw}$, where the original solver fails in the hot, low-pressure corner), on a grid of $X = \ln T$ and $Y = \ln P$, and stored in a table. For every node the table holds the three functions

$$\ln \rho(X,Y), \qquad \ln n_e(X,Y), \qquad \ln \chi_{500}(X,Y),$$

together with their derivatives ∂_X, ∂_Y and $\partial_X\partial_Y$. Between the nodes each function is evaluated by **bicubic Hermite interpolation**: a cubic polynomial in each direction that matches the stored values and derivatives at the four corners of the cell. The result is continuous with continuous first derivatives, which is important for the Newton-type solvers of the magnetohydrostatic modes, and it provides the derivative $\partial \ln\rho/\partial \ln P|_T$ that they need. The table used by all modes is `cache/eos_table_fullz.h5`. Its range of T and P (its *envelope*) covers the input atmospheres; a point outside the envelope is reported as an equation-of-state failure, never extrapolated.

Two derived quantities appear repeatedly:

$$q \equiv \frac{\rho}{P} \quad [\mathrm{s^2\,m^{-2}}], \qquad H \equiv \frac{P}{\rho\, g} = \frac{1}{g\, q} = \frac{k_B T}{\bar\mu\, m_{\rm u}\, g} \quad [\mathrm{m}]. \tag{6}$$

H is the **pressure scale height**. For $T = 6000$ K and $\bar\mu = 1.3$, $H \approx 140$ km.

5. The equation solved

5.1 Logarithmic form

Divide Eq. (2) by P and use $u = \ln P$ (so $du = dP/P$) and $\rho = q\,P$:

$$\boxed{\frac{du}{dz} = -g\, q\big(T(z),u\big) = -g\, \exp\big(\ln\rho(X,u) - u\big) = -\frac{1}{H(T,P)}} \tag{7}$$

In an isothermal gas of constant $\bar\mu$, H is a constant and the solution is $P = P_0\, e^{-(z-z_0)/H}$: the pressure falls by a factor e every scale height. Integrating in u instead of P has two advantages: a column spans eight or more orders of magnitude in P, but u changes smoothly and by moderate amounts per grid interval, which is what a numerical integrator handles best; and $P = e^u$ is positive whatever happens.

5.2 The boundary condition: one pressure per column

Equation (7) is a first-order ordinary differential equation for $u(z)$ in each column. It fixes how the pressure changes with height, not its value: if $u(z)$ is a solution, a different starting value gives a different, equally valid solution. One value per column is needed, and the mode prescribes the **same pressure $P_{\rm ref}$ at the same level in every column**:

$$u(x_i, y_j, z_a) = \ln P_{\rm ref} \qquad \text{for all columns } (i,j), \tag{8}$$

where the anchor level z_a is chosen by `column_hse.anchor`:

- `bottom` (default): $z_a = z_{N_z}$, the lowest level; the integration runs upward.
- `top`: $z_a = z_1$; the integration runs downward ($P_{\rm ref}$ defaults to 0.1 Pa if not set).
- `height`: $z_a = z_{\rm ref}$ anywhere inside the grid; the integration runs upward and downward from it.

$P_{\rm ref}$ is either a number in the configuration or, with `P_ref = "from_input"`, the attribute `P_ref_Pa` of the input file. For the production input this attribute is the median of the simulation's own pressure over the bottom plane.

Why the bottom? The anchor condition (8) declares the anchor plane to be a surface of constant pressure. Deep in the atmosphere, below the visible surface, the gas is dense, the relative temperature differences between columns are smaller and the true pressure varies much less across a horizontal plane than higher up, so this is the level at which the assumption is least wrong. All differences between the columns above it then come from their temperature profiles. With a top anchor the photospheric pressure of each column would instead depend on the hot, strongly structured chromosphere and transition region above it.

Existence and uniqueness. The right-hand side of Eq. (7) is a smooth function of z and u (the table interpolation has continuous first derivatives), so for every column the equation with its anchor value has exactly one solution, as long as (T,P) stays inside the table.

5.3 Statement of the problem

For every column (i, j), $i = 1 \dots N_x$, $j = 1 \dots N_y$, independently: *find* $u(z)$ *on* $z_{N_z} \le z \le z_1$ *such that* $du/dz = -g\,q(T, u)$ *and* $u(z_a) = \ln P_{\rm ref}$*; then compute* ρ, n_e, χ_{500} *from the table, the column mass and the optical depth.* There is no coupling between columns: the problem consists of $N_x N_y$ independent one-dimensional initial-value problems.

6. Assumptions

A1. Static gas. All velocities are neglected; the momentum flux and the pressure of turbulent or wave motions are not included.
A2. No magnetic force. The field, if present in the input, is not used.
A3. Vertical balance only, column by column. Only the z component of $\mathbf{F} = 0$ is imposed; the horizontal components are left unbalanced (Sect. 10).
A4. Equal pressure on the anchor plane. Every column has $P = P_{\rm ref}$ at z_a (Sect. 5.2). This is the only external pressure information.
A5. Constant, vertical gravity. g is the same everywhere; the curvature of the Sun and the decrease of g with height (0.4 % over 3 Mm) are neglected (plane-parallel geometry).
A6. LTE ideal gas of solar composition, with ionisation and molecules from the Saha and molecular equilibria (Sect. 4).
A7. Temperature between grid levels: $\ln T$ varies linearly with z inside each grid interval.
A8. Optical depth: continuum extinction at 500 nm along the vertical (or along a fixed angle with $\mu \neq 1$), without lines; the gas above the grid is represented by an exponential extrapolation of χ_{500} (Sect. 8).

7. Numerical solution

7.1 Temperature inside an interval

The temperature is known only on the grid levels. Inside the interval $[z_{k+1}, z_k]$ the integrator needs it at arbitrary heights; by assumption A7,

$$X(z) = X_{k+1} + (X_k - X_{k+1})\,\frac{z - z_{k+1}}{z_k - z_{k+1}}, \qquad X = \ln T. \tag{9}$$

Interpolating $\ln T$ rather than T keeps the temperature positive and represents an exponential rise (such as the transition region) better.

7.2 The classical Runge–Kutta method

Write Eq. (7) as $du/dz = f(z, u)$ with $f = -g\,q(X(z), u)$. A step of length s from (z, u) evaluates the slope four times,

$$\begin{aligned} k_1 &= f(z,\, u), & k_2 &= f\big(z + \tfrac{s}{2},\, u + \tfrac{s}{2}k_1\big), \\ k_3 &= f\big(z + \tfrac{s}{2},\, u + \tfrac{s}{2}k_2\big), & k_4 &= f(z + s,\, u + s\,k_3), \end{aligned} \qquad u(z+s) \approx u + \frac{s}{6}\,(k_1 + 2k_2 + 2k_3 + k_4). \tag{10}$$

The slopes at the beginning, twice in the middle and at the end are averaged with weights $1, 2, 2, 1$ (Simpson's rule). The error per step is proportional to s^5 and the total error to s^4: halving the step reduces the error by about a factor 16. For an upward integration $s > 0$, for a downward one $s < 0$.

7.3 Substeps and their control

Each grid interval is crossed in n equal substeps. The accuracy is controlled interval by interval:

1. Integrate the interval with $n = n_{\rm sub}$ substeps (default 4) and with $2n$ substeps, from the same starting value.
2. If the two end values differ by more than `rk_tol` (10^{-6} in u, i.e. a relative pressure difference of 10^{-6}), double n and compare again ($2n$ against $4n$, …), up to `n_sub_max` = 64 substeps.
3. Keep the result with the larger number of substeps.

Because the error falls by a factor 16 per doubling, the difference between two successive results is a good estimate of the error of the coarser one, and the kept, finer result is more accurate than the tolerance. Steep temperature changes, such as the transition region, automatically receive more substeps. Two flags are stored per column: whether any interval needed refinement, and whether any interval still missed the tolerance at 64 substeps.

7.4 Order of the integration for the three anchors

- `bottom`: $u_{N_z} = \ln P_{\rm ref}$; then intervals $N_z \to N_z - 1 \to \cdots \to 1$ (upward).
- `top`: $u_1 = \ln P_{\rm ref}$; then $1 \to 2 \to \cdots \to N_z$ (downward).
- `height`: find the interval $[z_{k_r+1}, z_{k_r}]$ that contains $z_{\rm ref}$; integrate from $z_{\rm ref}$ up to z_{k_r} and on upward, and from $z_{\rm ref}$ down to z_{k_r+1} and on downward.

At the anchor level the stored pressure is exactly $P_{\rm ref}$. If a substep reaches a (T, P) outside the table, the column is marked as failed and its outputs are left undefined (NaN).

7.5 Column mass

Integrating Eq. (2) from a height z to the top of the grid gives

$$P(z) = P(z_1) + g \int_z^{z_1} \rho \, dz' \equiv P(z_1) + g\, m(z) : \tag{11}$$

the pressure at a point equals the weight of the gas above it per unit area, plus whatever presses on the top of the grid. The code integrates $dm/dz = -\rho$ with the same Runge–Kutta substeps as u and stores $m(z_k)$ counted from the top plane. It also stores an independent trapezoidal sum $\sum \frac{1}{2}(\rho_k + \rho_{k+1})(z_k - z_{k+1})$ on the grid levels. The two agree closely, and $g\,m$ should match $P - P(z_1)$; both serve as checks of the integration.

7.6 Derived quantities

At every grid point, with X_k and u_k known, the table gives $\ln\rho$, $\ln n_e$ and $\ln\chi_{500}$. From them $P_e = n_e k_B T$, $\bar{\mu} = \rho k_B T/(P m_{\rm u})$ and $n_{\rm H} = \rho/(\text{mass per hydrogen nucleus})$, the latter being fixed by the composition.

8. Optical depth

8.1 Definition

Light of intensity I crossing a path ds loses $dI = -\chi I\, ds$, so $I = I_0 e^{-\tau}$ with $\tau = \int \chi\, ds$, the number of absorption lengths crossed. For light leaving vertically the optical depth of height z is

$$\tau_{500}(z) = \tau_{\rm top} + \int_z^{z_1} \chi_{500}(z')\, dz', \tag{12}$$

which grows downward. For a viewing angle θ the path through each layer is longer by $1/\mu$, $\mu = \cos\theta$, and the stored value is τ/μ (default $\mu = 1$). Light emitted at optical depth τ escapes with probability $e^{-\tau}$, so most of the continuum light we see comes from around $\tau_{500} \approx 1$, the "visible surface".

8.2 Discretisation

The integral is evaluated with the trapezoidal rule on the grid levels, starting from the top:

$$\tau_1 = \tau_{\rm top}, \qquad \tau_{k+1} = \tau_k + \tfrac{1}{2}\,(\chi_k + \chi_{k+1})\,(z_k - z_{k+1}), \quad k = 1 \ldots N_z - 1. \tag{13}$$

8.3 The gas above the grid

The atmosphere continues above z_1. If χ_{500} keeps falling exponentially there, $\chi(z) = \chi_1 e^{-(z-z_1)/H_\chi}$, the optical depth of everything above the grid is

$$\tau_{\rm top} = \int_{z_1}^{\infty} \chi_1\, e^{-(z-z_1)/H_\chi}\, dz = \chi_1 H_\chi, \qquad H_\chi = \frac{z_1 - z_2}{\ln(\chi_2/\chi_1)}, \tag{14}$$

with H_χ measured from the top two levels. This is used (option `exp_tail`, the default) when it makes sense: $\chi_2 > \chi_1 > 0$ (the extinction decreases upward) and $0 < H_\chi < 5000\,$km. Otherwise half a grid cell is used, $\tau_{\rm top} = \frac{1}{2}\chi_1(z_1 - z_2)$. The options `half_cell` and `zero` force the alternatives.

8.4 The height of $\tau_{500} = 1$

In every column the code finds the two levels with $\tau_k \le 1 \le \tau_{k+1}$ and interpolates linearly in $\log_{10}\tau$:

$$z_{\tau=1} = z_k + (z_{k+1} - z_k)\,\frac{0 - \log_{10}\tau_k}{\log_{10}\tau_{k+1} - \log_{10}\tau_k}. \tag{15}$$

9. Iteration and acceptance criteria

There is no global iteration in this mode: every column is integrated once, and the result is the solution of the discretised initial-value problem. The only accuracy criteria are local:

- per grid interval, $|u^{(2n)} - u^{(n)}| \le 10^{-6}$ (the Runge–Kutta tolerance, Sect. 7);
- every evaluated (T, P) lies inside the table envelope.

The run reports how many columns needed refinement, how many missed the tolerance, the largest number of substeps used, how many columns used each $\tau_{\rm top}$ closure, and how many columns left the table. If any column left the table the run ends with the status `eos_failure`; otherwise it is `converged`. Because no reductions over columns enter the values, the result is identical for every number of processes and threads.

10. Three dimensions: what the result does not satisfy

10.1 The residual force

After the integration every column obeys the z component of Eq. (1). The full three-dimensional imbalance is evaluated on the grid as a diagnostic,

$$F_x = -\frac{\partial P}{\partial x}, \qquad F_y = -\frac{\partial P}{\partial y}, \qquad F_z = -\frac{\partial P}{\partial z} - \rho g, \tag{16}$$

with three-point, second-order finite differences (x and y periodic, z on the non-uniform grid). The layer medians and 99th percentiles of $|F_h|/(\rho g)$ (with $F_h = \sqrt{F_x^2 + F_y^2}$) and of $|F_z|/(\rho g)$ are reported: the unbalanced force in units of the weight of the gas. F_z is small (it measures only the difference between the Runge–Kutta solution and a three-point derivative on the grid); F_h is not.

10.2 Why horizontal balance is impossible without help

Consider two neighbouring columns at the same height, one hot and one cool. In the hot column the scale height is larger, so above the common anchor plane the pressure falls more slowly: at the same height the hot column has a higher pressure. There is a horizontal pressure gradient that nothing balances. Can *any* pressure field make a non-magnetic atmosphere static? Take the curl of $\mathbf{F} = 0$. The curl of a gradient is zero, and $\nabla\times(\rho g\hat{\mathbf{z}}) = g\,(\partial_y\rho,\, -\partial_x\rho,\, 0)$, so

$$\frac{\partial \rho}{\partial x} = \frac{\partial \rho}{\partial y} = 0. \tag{17}$$

A static, non-magnetic atmosphere must have a uniform density on every horizontal plane, and then also a uniform pressure. A structured atmosphere (hot granules beside cool intergranular lanes) cannot be static: in the Sun the imbalance drives flows, and in a static model only a magnetic force could take its place. The magnetohydrostatic modes of HSEMHS address this; the present mode deliberately does not, and labels its output *vertical HSE only*.

11. Limitations

- The horizontal force balance is not satisfied (Sect. 10); the result is not a three-dimensional equilibrium.
- The absolute pressure level rests entirely on $P_{\rm ref}$ and on the assumption that the anchor plane is isobaric. A different $P_{\rm ref}$ shifts $\ln P$ of a column by nearly the same amount at all heights (exactly so for constant $\bar{\mu}$), and moves the τ scale accordingly.
- Dynamic pressure, magnetic pressure and tension are absent, so in strongly magnetised or fast-moving regions the pressure can differ substantially from that of the original simulation.
- The optical depth uses continuum extinction only and a model of the gas above the grid.

12. Summary of the algorithm

1. Read T, x, y, z; set g and $P_{\rm ref}$; load the equation-of-state table.
2. For every column independently (in parallel): set $u = \ln P_{\rm ref}$ at the anchor; integrate $du/dz = -g\rho/P$ and $dm/dz = -\rho$ interval by interval with RK4, $\ln T$ linear in z, substeps doubled until two successive results agree to 10^{-6} (at most 64).
3. At every level evaluate ρ, n_e, χ_{500} from the table; derive P_e, $\bar{\mu}$, $n_{\rm H}$ and the trapezoidal column mass.
4. Integrate τ_{500} from the top with the trapezoidal rule, starting from $\tau_{\rm top} = \chi_1 H_\chi$; find $z(\tau_{500} = 1)$.
5. Evaluate the three-dimensional residual force as a diagnostic; write all fields and the per-layer statistics.